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April 30, 1974
SUPERSEDING
Fed. Spec. S-O-865A
March 26, 1969

FEDERAL SPECIFICATION

OVENS, REVOLVING TRAY, BAKING AND ROASTING, COMPACT (GAS, ELECTRIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers small, commercial revolving tray ovens of the compact type (see 6.1).

1.2 Classification. The revolving tray ovens covered by this specification shall be of the following types, sizes, styles, and models as specified (see 6.2).

Type I - Direct electric heating system.
Type II - Direct gas heating system.
  Model a - For natural gas.
  Model c - For liquid petroleum gas (LP).

Style A - Oven door on left side.
Style B - Oven door on right side.

Size 6 - 6 tray.
Size 7 - 7 tray.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 7310

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago Kansas City, MO, Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

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Military Specifications:

MIL-B-2316	-	Bakery Equipment, Including Units, Assemblies, Repair Parts and Tools, Preparation for Delivery Of.
MIL-S-4174	-	Steel Sheet and Strip, Flat, Aluminum Coated, Low Carbon.

Military Standards:

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Electrical Manufacturer's Association (NEMA)

Standard MG1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

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Underwriters' Laboratories Inc., (UL)

Standard UL73 - Motor Operated Appliances

Standard UL197 - Commercial Electric Cooking Appliances

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, LI, NY 11746; or 1655 Scott Blvd., Santa Clara, CA 95050.)

American National Standards Institute (ANSI)

Standard No. Z50.1 - American Standard Safety Code for Baking Equipment

Standard No. Z21.28 - Commercial Gas Baking and Roasting Ovens

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10017.)

American Gas Association (A.G.A.)

Directory of Certified Appliances and Accessories

(Application for copies should be addressed to the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44141.)

Baking Industry Sanitation Standards Committee (BISSC)

Standard No. 14 - Mechanical Ovens

(Application for copies should be addressed to the Baking Industry Sanitation Standards Committee, 54 Fifth Avenue, New York, NY 10017.)

National Sanitation Foundation (NSF)

Standard No. 4 - Gas and Electric Commercial Cooking and Warming Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

American Society for Testing and Materials (ASTM)

Designation A167 - Stainless Chromium-Nickel Steel Plate, Sheet, and Strip

Designation A176 - Stainless Chromium Steel Plate, Sheet, and Strip

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.2 and 6.4).

3.2 Standard product. The oven shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the government consistent with their normal commercial practices.

3.3 Codes and standards. All ovens shall comply with the applicable requirements of ANSI Standard No. Z50.1, Standard UL197, and NSF Standard No. 4 or BISSC Standard No. 14. Gas-fired ovens shall comply with ANSI Standard No. Z21.28. Electric motors shall comply with the requirements of NEMA Standard MG1.

3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the ovens he proposes to furnish under this specification meet the applicable requirements of ANSI Standards Z50.1 and Z21.28; Standard UL197 and NSF Standard No. 4 or BISSC Standard No. 14.

3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label on the oven (or the UL label on each of the major electrical components of the oven), a UL listing mark, or a certified test report from a recognized testing laboratory acceptable to the Government, indicating that the ovens (or the electrical components of the ovens) meet the applicable requirements of Standard UL197.

3.3.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

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(1) Listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished oven, or

(2) A certification for the oven issued by NSF under their special one-time evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the ovens are being procured, indicating that the ovens have been tested and conform to NSF Standard No. 4.

3.3.1.3 BISSC. Acceptable evidence of meeting the requirements of BISSC shall be:

(1) A photostatic copy of the current BISSC certificate of authorization for the oven, or

(2) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the ovens are being procured, indicating that the units have been tested and conform to the applicable requirements of BISSC Standard No. 14.

3.3.1.4 A.G.A. Acceptable evidence of meeting the terms of certification of A.G.A. shall be:

(1) A photostatic copy of the A.G.A. Appliance Certificate.

(2) A listing of the oven (or the gas accessories of the oven) in the A.G.A. Directory of Certified Appliances and Accessories.

(3) A certified test report from a recognized independent testing laboratory acceptable to the Government stating that the ovens have been tested and conform to the applicable requirements of ANSI Standard No. Z21.28.

3.4 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial oven, provided the completed item complies with all provisions of this specification.

3.4.1 Stainless steel sheets. Stainless steel sheets shall conform to any of the type 300 series of ASTM A167, or any of the type 400 series of ASTM A176.

3.4.2 Aluminized steel sheets. Aluminized steel sheets shall conform to type 1, grade A of MIL-S-4174.

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3.5 Design. Revolving tray ovens shall be designed for general baking and roasting. The tray surface shall be designed to support a uniform load of not less than 20 pounds per square foot of tray area, and shall not bend or show permanent deformation when tested as specified in 4.7.2.1. The oven chamber shall be heated, vented, and otherwise designed to provide uniform baking during continuous or intermittent tray rotation. The lateral heat distribution and top and bottom heat balance shall be regulated to eliminate hot spots. The oven temperature shall not drop more than 40 degrees Fahrenheit (F), shall recover to 425° F within 15 minutes, and shall satisfactorily bake pies when tested as specified in 4.7.1. The type I oven heating system shall raise the temperature of the oven 375° F above ambient in a maximum of 1 hour and the type II oven heating system shall raise the temperature of the oven 375° F above ambient in a maximum of 45 minutes, when tested as specified in 4.7.2. The size 6 oven shall be provided with 6 trays, and the size 7 oven with 7 trays. Type I (electric) ovens shall have no controls on the oven side away from the door. The left sidewall of the oven with the door on the right (style B) shall be designed to be placed flush against the right sidewall of the oven with the door on the left (style A).

3.6 Construction. The oven shall be constructed so that the load of the reel assembly, loaded trays and drive assembly, shall be transmitted through structural framework directly to the foundation. The oven shall consist of preinsulated, self-supported panels with corner frames, or panels insulated after erection. Panels shall be properly reinforced and welded or bolted into sections for convenient handling in assembly. Panels shall be so constructed that heat transfer from the inner lining to the outer shell is reduced to a minimum. The base shall be welded to form a single plate and shall be fabricated from steel sheet not thinner than 10 gage or the base shall be welded from structural steel. The sides shall be plumb and square with the base. The entire construction shall be so engineered that after being operated at 500° F for 3 hours, expansion and contraction of oven panels shall not cause twisting, bulging, or permanent distortion (see 4.7.3). The size 6 oven shall be provided with six trays each having clear dimensions of at least 21 inches by 26-1/2 inches. The size 7 oven shall be provided with seven trays and shall have a capacity of seven bun pans having the dimensions of at least 18-1/8 inch by 26-1/4 inches. The overall dimensions of the oven shall be not more than 90 inches wide by 45 inches deep by 88 inches high.

3.6.1 Inner walls, top, and bottom. The inner walls, top, and bottom of each oven shall be constructed of aluminized steel sheets specified in 3.4.2, not thinner than 20 gage. The aluminized sheets shall be spot welded or bolted to the steel framework to form continuous unbroken panels. Sheared edges of the aluminized sheets shall be arranged on the framework to obviate exposure to the oven interior, or shall be treated to resist corrosion in accordance with the manufacturer's standard practice. The wall and top sections shall be flanged or lapped for bolted assembly. Asbestos gaskets and cement shall be used between all joints, and bolts shall be spaced to insure steam-tight joints.

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3.6.2 Outer walls and top.

3.6.2.1 Top cover and rear wall. Unless only aluminized steel is specified (see 6.2), the top cover and rear wall shall be fabricated from aluminized or galvanized steel. The rear wall shall be not thinner than 20 gage.

3.6.2.2 Sides and front wall. Outer sides and front wall shall be fabricated from stainless steel not thinner than 20 gage and the exterior surface shall be protected from tool marks during fabrication with the use of gummed paper or spray coating. Protection material shall not be removed from the panel until final erection of the oven is completed and it is ready for firing. Metal sheets shall be smooth, free of dents, ripples, or scratches and secured to the frame without exposed fastening devices such as nuts, bolts, or screws.

3.6.3 Insulation. The thermal conductivity of the insulation shall be such that with an oven temperature of 500° F for 3 hours, the outside walls and top surface temperature shall not exceed 60° F above ambient when tested as specified in 4.7.3.

3.6.4 Doors.

3.6.4.1 Oven door. The oven door shall be of the preinsulated, panelized construction, or shall be constructed of 3/16-inch aluminum plate or stainless steel not thinner than 10 gage. The interior of the panelized door shall be fabricated from aluminized steel having a finished thickness of not less than 18 gage. The exterior of the door shall be fabricated from stainless steel having a finished thickness not thinner than 20 gage. The door shall be hinged at the bottom, opening outward so the inner surface serves as a loading shelf when in the opened position. The door shall be provided with a cool-type, plastic composition or hardwood handle not less than 10 inches in length. The door shall have a heat resisting glass observation panel of not less than 36 square inches. Means shall be provided to facilitate replacement of the glass in the event of breakage. The door opening shall be at least 12 inches high by 22 inches wide, centered with the tray.

3.6.4.2 Clean-out and access doors. Each oven shall be provided with a clean-out door or access door, or removable panel, to the oven chamber, combustion chamber, and drive machinery compartment. Access panels shall be at least 18 inches wide by 20 inches high. The access door(s) of type I ovens (electric) shall be located at the front of the oven or at the left side of style A ovens or on the right side of style B ovens. Access door(s) for type II ovens shall be located at the front of the oven. Doors shall be insulated and shall allow inspection, maintenance, and replacement of heating elements or gas equipment, as applicable.

3.6.5 Reel mechanism.

3.6.5.1 Spiders. Spiders shall be fabricated from steel and reinforced, welded, and machined to insure free unbinding operation. Spokes of spiders shall be steel

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on the main shaft to prevent distortion and misalignment, and shall be designed to carry the trays in a level position at all times. Tray carriers shall be designed to permit removal of the trays through the loading door.

3.6.5.2 Reel shafts. The reel shall consist of two spiders joined to a steel main shaft. One spider shall be located at each end of the shaft. Entire reel mechanism shall be mounted on the main shaft which shall extend throughout the entire depth of the oven. Journals of reel shaft shall be designed to be integral with the shaft and shall be of a size and rigidity to avoid undue bearing wear.

3.6.5.3 Bearings. Main shaft bearings shall be mounted outside the heat zone, ventilated from the exterior, and shall be of the self-aligning, pillow block design. The main bearings shall be of the ball or roller bearing type with means for readily accessible lubrication, when the oven is backed up against a wall except the rear main bearing may be a self-lubricating, solid metalized carbon graphite type, in which case lubrication means for the rear bearing is not required. All bearings inside the oven shall be permanently lubricated.

3.6.6 Trays. The trays shall be of a solid type or, when specified (see 6.2), grill type using expanded metal or spaced bars. Trays shall be rigid, of welded steel construction, braced with steel angles or similar structural shapes to prevent distortion under severe conditions of uneven loading. Baking or roasting surfaces shall be not thinner than 10 gage steel plate fastened to the tray framework to permit free, unbinding expansion. All trays shall have a vertical lip at sides and rear, at least 4 inches high, and perforated if the trays are grill type. Trays shall be attached to the tray shaft in a manner that will permit leveling adjustments for each tray. The trays shall be stabilized by means of stabilizer ring, guide wheels, and tray arms. The stabilizer ring shall be the same diameter as the spiders.

3.6.6.1 Tray frames. When specified (see 6.2), a tray frame shall be provided for each tray of the size 6 (6 tray) oven to double the capacity for certain specialized uses. The frame shall slide onto a tray and accommodate two 18 inch by 26 inch bun pans, one above the other or four 12 inch by 20 inch steam table pans. The tray frame shall be constructed so the bun pans are evenly spaced within the door opening when the frame is installed on the tray. The top of the frame shall be inclosed to prevent baking products from hitting other trays or parts of the oven when the oven is in operation. Tray dimensions for 6 tray and 7 tray ovens shall be at least 18-1/2 inches by 26-1/4 inches when a tray frame is specified.

3.6.7 Drive. The oven transmission unit shall consist of a soft torque start, ball bearing motor, a speed reducer unit equipped with ball bearings,

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sprocket, and roller chain connection from speed reducer to the reel sprocket. No drive transmission components requiring lubrication shall be inside the heated zone. V-belt pulley on the motor shall be of a type that will maintain proper tension at all times. Entire transmission unit shall be designed to prevent coasting or back lash of the reel under unbalanced load conditions and shall be arranged to maintain proper drive chain tension at all times. Drive motor shall be the reversing type, capable of rotating the reel clockwise and counter clockwise with equal performance, and shall be controlled by means of a switch located at the front of the oven accessible from the loading door. Means shall be provided for manual operation of the trays in case of temporary current failure. A hand-crank, operable from the front of the oven, shall be provided for manual rotation of the trays in case of power interruption. The force required for tray-rotation speeds equal to, or greater than, when motor-driven, shall not exceed 10 pounds. An override shall be provided to protect the operator in case power is restored during manual operation. Drive motor shall be provided with overload protection. A device shall be provided in each oven drive to prevent the possibility of internal damage to the oven or pans due to jamming. The device shall shut off the drive motor and sound a continuous or intermittent alarm. Means shall be provided for manually shutting off the alarm and releasing the jam (see 4.7.4).

3.6.8 Ventilation and flues. Ovens shall be provided with proper ventilation of the combustion gases and cooking fumes. Venting from the oven chamber shall be accomplished from a point below the level of the oven loading door and shall be balanced so no special dampers are required. A ventilating hood shall be provided over the oven door opening and shall be concealed within the front panels of the oven. The hood shall extend across the entire width of the oven loading door. Ventilating hood shall operate when the oven door is opened and shall be designed to prevent hot gases from reaching the oven operator when tested as specified in 4.7.4.1. Hood outlets and oven vents shall be carried to the top of the oven.

3.6.9 Heating system. Heating system shall be designed for direct heating and shall be arranged to provide uniform distribution of heat throughout the baking chamber.

3.6.9.1 Type I. The electrically heated ovens shall be provided with electric oven heating units with a minimum input of 16 kilowatts. Unless otherwise specified (see 6.2), the heating unit shall operate on a nominal 208-volt, 60-hertz, 3-phase, alternating current, power source. The heating

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elements shall be compacted tubular sheath type to preclude short circuiting or mechanical injury in the event heavy metal objects or other conductible materials fall onto the heating unit. Entire heating unit shall consist of a series of elements arranged for easy removal from the oven for repair or replacement of individual elements or the entire heating unit. The temperature in the oven shall be controlled automatically by an adjustable themostat having a range of 100° F to 550° F and capable of maintaining a uniform temperature across each tray that shall not vary more than 5° F when tested as specified in 4.7.2. A corrosion-resistant switch box, housing the switch, thermostat contactor, cut-outs, and wiring, shall be located outside the heat zone of the oven and shall be accessible for inspection and repair. There shall be a disconnect switch permitting entry to the switch box only after the circuit is opened.

3.6.9.2 Type II. The gas burner of the type II, model a oven shall be for use with natural gas and for the type II, model c oven shall be for liquefied petroleum (LP) gas. Each gas burning system shall be provided with a fully automatic temperature control, safety pilot, flame protection device controlling the main burner, and a signal light visible from the front of the oven to indicate operation of the main burner. Gas burners shall be stainless steel insert type, extending the full length of the trays. There shall be an automatic gas valve control and flame protection device. The flame failure system controlling the main gas supply shall prevent the flow of gas, and sound a continuous alarm within 5-seconds of flame failure, when tested as specified in 4.7.4. Covering the entire burner flame shall be a removable steel baffle to protect the burner parts from falling food particles and to protect tray bottoms from direct contact with the flame.

3.6.10 Motor and wiring. Unless otherwise specified (see 6.2), the motors for all ovens shall operate on a nominal 120-volt, 60 hertz, 1-phase, alternating current power system, or a 208-volt, single phase jumper for type I ovens. All motors shall conform to NEMA Standard MG1. The supplier shall furnish all controls, interconnecting wiring, conduit, outlet boxes, connectors, and related items that are required for operation of the oven. The oven shall be prewired at the factory ready for connection by others.

3.6.11 Accessories. Each oven shall be provided with the following accessories:

3.6.11.1 Tray indicator. A dial type tray indicator with a minimum diameter of 5 inches shall be provided at the front of the oven. The tray indicator shall be synchronized with the oven reel to designate the position of any trays inside the oven, and shall have a means for indicating which tray is located at the loading door.

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3.6.11.2 Interior illumination. The oven shall be provided with interior lighting located to provide adequate overall interior illumination. The illuminators shall be of the marine type, with gasketed, steamlight globes, ventilated and replaceable from outside the oven chamber.

3.6.11.3 Clock. The oven shall be provided with a constant-ringing timing clock with signal bell, which can be set for baking and roasting periods up to 3 hours.

3.6.11.4 Steam injection system. When specified (see 6.2), each oven shall be equipped with a manually operated steam spray system of not less than one spray arm, located near the top of the oven chamber in a manner that will insure equitable application of steam over the products being baked. The steam control valve shall be situated in a readily accessible location for the operator. A steam trap shall be furnished to collect condensate. The entire steam injection system shall be complete to the steam supply service connection. All parts of the steam injection system shall work properly, and the steam injected from the nozzles shall be free of visible water droplets when tested as specified in 4.7.5.

3.6.11.5 Indicating dial thermometer. When specified (see 6.2), an indicating-dial thermometer shall be provided on the front of the oven with a chart recorder.

3.7 Repair and maintenance. Provisions shall be made for ready adjustment, service, and replacement of all components of the reel mechanism and drive, heating system, lighting system, and controls without disturbing or disassembling any unrelated major component or equipment. In addition, there shall be free access to inspect or service and adjust any other component or equipment, as required, for proper operation of the ovens.

3.8 Lubrication. Bearings and other parts requiring pressure lubrication shall be provided with pressure-type grease fittings. At the time of delivery, the oven shall be completely lubricated with the manufacturer's standard lubricant.

3.9 Marking for identification. Marking for Military agencies shall be applied in accordance with MIL-STD-130. Marking for Civilian agencies shall be in accordance with the manufacturer's standard practice.

3.10 Workmanship. The finished revolving tray ovens shall be clean and well made.

3.10.1 Steel fabrication. Steel used in the fabrication of the ovens shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping

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shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.10.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers shall be provided in accordance with commercial practice, and all bolts, nuts, and screws shall be drawn tight.

3.10.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, and in full contact with the surfaces of the riveted members.

3.10.4 Welding and brazing. All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished to prevent the occurrence of burn holes, cracks, fractures, or incomplete fusion. All scale or flux deposits shall be removed from finished welds.

3.11 Electromagnetic compatibility. When specified (see 6.2), revolving tray ovens for Military use shall be designed and equipped for electromagnetic compatibility in accordance with IIB of MIL-STD-461 (see 4.7.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a first article is required, the oven shall be examined for defects in table I, dimensions specified, and subjected to the tests in column "b" of table II. The presence of any defect, or the failure of any test shall be cause for rejection of the first article.

4.3 Inspection.

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4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process (and all items processed) as follows:

a. Sheet metal. Check sheet metal material and gages for compliance with the requirements of 3.6.

b. Corrosion treatment. Check anti-corrosion treatment for edges of sheared sheets and spot welds (see 3.6.1).

c. Check to see that exterior surfaces are protected during fabrication (see 3.6.2.2).

4.3.3 End-item inspection.

a. Set-up of the completely fabricated oven at the contractor's plant will not be required.

b. Final inspection (examination and tests) of the completely fabricated end item shall be performed by the contractor after installation of the oven at destination, provided the contractor is required to supervise the installation of the oven at destination (see 6.2 and 6.3).

c. In the event the contractor is not required to supervise the installation of the oven at destination, the requirement for the contractor to test the complete end item will be waived provided the contractor can certify that the item has been in commercial or military use for not less than 1-year and that there are no reported problems with the item. However, the government reserves the right to inspect and test the equipment after installation and to reject defective equipment.

d. The procuring activity will specify the government installation where the end item inspection is to be conducted (see 6.2 and 6.3).

4.3.3.1 Visual and dimensional examination. Examination of each oven shall be made for visual defects in table I and for dimensions specified. Failure to make corrections whenever a deviation to specified requirements is noted shall be cause for rejection of the oven. Correction records shall be kept available for review by the Government representative.

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TABLE I. Classification of visual and dimensional defects

Examine	Defect
Finish	Finish not as specified Not finished where required Indication of rust
Construction and workmanship (general)	Any component missing or not type or size specified Any component fractured, split, punctured, dented, bowed, or malformed Sharp burr, sliver, or splinter Any functioning component that requires abnormal force to operate Bearing inside the heated zone not permanently lubricated type Any part misplaced, loose, or not in proper alignment Any adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended Any component not readily accessible for servicing (where specified)
Dimensions	Not as specified
Oven doors and door handle	Not provided with a cool type plastic composition or hardwood handle
Ventilation and flues	Not provided with stack damper (when applicable) Ventilating hood not provided over the oven door openings or not concealed within the front top panel of the oven
Reel mechanism	Spider does not carry the trays in a level position at all times Tray carriers do not permit easy removal of trays Tray not provided with levelizer
Welding	Missing, incomplete, burn holes, cracked, fractured, or not fused Slag inclusion, undercut, not smooth or uniform Poor surface appearance of weld as from craters, piling up of deposited metal, evidence that weld has not been ground or filed where required, or grinding or filing not neatly accomplished

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Examine	Defect
Bolts, nuts, screws, studs and other types of fasteners	Missing, broken, stripped, fractured, or loose
Drive motor and heating system	Not type of design specified Does not function as specified
Gas burning equipment (if applicable)	Any accessory not as specified
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal, or not enclosed in specific conduit where required Slack not provided for wiring to relieve strain or excessive insulation stripped from wiring
Lubrication (as applicable)	Not lubricated where required Not specified type lubrication fittings Lubrication fitting not readily accessible for servicing Drive transmission components requiring lubrication mounted inside the heated zone
Identification marking and name plate	Missing, incomplete, illegible

4.4 End item testing. When the supplier is required to supervise the installation of the oven at destination (see 4.3.3), the oven shall be tested in accordance with 4.7. Failure to pass any test shall be cause for rejection of the oven. When specified (see 6.2), the Government will furnish the baking products for the baking test (see 4.7.1), when the tests are conducted in Government installations within continental United States.

4.5 Inspection of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing, and marking required by MIL-B-2316, are complied with.

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4.6 Certificate examination. Certificates of compliance, certified test reports, approval labels, or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements, shall be examined and validated.

4.7 Test methods. The test schedule shall be in accordance with table II:

TABLE II. Test schedule

Test seq.	First article	Each oven	Test	Test para.	Req. para.
a	b	c	d	e	f
1	X	X	Baking, temperature drop, and recovery	4.7.1	3.5
2	X	X	Performance	4.7.2	3.5 3.6.9.1
3	X	X	Tray load	4.7.2.1	3.5
4	X	X	Insulation	4.7.3	3.6 3.6.3
5	X	X	Safety	4.7.4	3.6.7 3.6.9.2
6	X	X	Door vent	4.7.4.1	3.6.8
7	X	X	Steam apparatus	4.7.5	3.6.11.4
8	When spec.	When spec.	Electromagnetic compatibility	4.7.6	3.11

4.7.1 Baking, temperature drop, and recovery test. Bring the temperature of the oven to 450° F and load the oven to capacity with 9 inch two crust cherry filled pies (see 6.5), which have a temperature between 55° F and 70° F. The pies shall be in parchment lined paper pie plates with metal rims and shall be loaded directly on the trays. During the test, check the temperature drop and recovery time. Inability of the oven to meet temperature drop, recovery time, and satisfactory pie-baking requirements of 3.5 shall constitute failure of this test.

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4.7.2 Performance tests. Starting with an empty oven at room temperature with the loading door closed, the oven shall heat up at the rate specified in 3.5. The temperature along any one tray shall not vary more than 5° F (see 3.6.9.1). Inability of the oven to heat up and maintain uniform temperatures as specified shall constitute failure of this test.

4.7.2.1 Tray load. Load each tray with a uniformly distributed load of 20 pounds per square foot and operate the oven for 10 minutes. Stop the oven and observe each tray for bends or permanent deformation to determine compliance with the requirements specified in 3.5. Inability of the tray to sustain the load as specified shall constitute failure of this test.

4.7.3 Insulation test. After the oven has been maintained at a temperature of 500° F for a minimum of 3 hours, the average temperature of the combined external surfaces of the heated enclosure (excluding the door, door frame, and access panel) shall be taken to determine conformance to 3.6.3. Not less than four temperature readings per surface, taken in the center of equal, uniformity distributed areas shall be used to compute the single, average, external temperature. During the course of the test, precautions shall be taken to prevent heat sources, such as other ovens, from increasing the external wall temperature of the oven being tested. After this test, oven panels shall be checked for twisting, bulging, or distortion to determine conformance with 3.6. Inability of the oven to meet thermal conductivity requirements shall constitute failure of the test.

4.7.4 Safety tests. The gas safety system shall be tested with the oven chamber at 200° F to 450° F and main gas burners and pilot ignited and operating. The oven thermostat shall be elevated (turned-up beyond registration of the indicating thermometer) to allow a longer operating time. Individual burner cocks shall be shut off and the pilot light extinguished to determine compliance with 3.6.9.2. The protective device against pan-jamming shall be checked on all type I and type II ovens to determine compliance with 3.6.7. Inability of the oven to meet flame failure and pan-jamming alarm requirements shall constiture failure of the test.

4.7.4.1 Door vent. After the oven has attained a temperature of 450° F. the oven door shall be opened by an operator standing in front of the door to determine compliance with the requirements specified in 3.6.8. Inability of the oven to keep hot gases away from the operator shall constiture failure of this test.

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4.7.5 Steam apparatus test. The steam apparatus, when required, shall be operated to determine compliance with 3.6.11.4.

4.7.6 Electromagnetic compatibility test. When electromagnetic compatibility is required, the first article or initial production unit, as applicable, shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Preservation, packaging, and packing for each level of protection shall be made in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The compact revolving tray ovens covered by this specification are used for modular perimeter installation in kitchens and pastry kitchens.

6.2 Ordering data. Purchasers should excercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Type, size, style, and model of oven to be furnished (see 1.2).
(c) When a first article is required (see 3.1, 4.2, and 6.4).
(d) When the top cover and rear wall are to be aluminized steel (for Navy and Marine Corps purchases) (see 3.6.2.1).
(e) When grill type trays are required (for V.A. purchases) (see 3.6.6).
(f) When tray frames are required for size 6 (6 tray) ovens (see 3.6.6.1).
(g) Oven electrical characteristics, if other than specified (see 3.6.9.1 and 3.6.10).
(h) When a steam injection system is required (see 3.6.11.4).

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(i) When an indicating dial thermometer is required (see 3.6.11.5).
(j) When electromagnetic compatibility is required (see 3.11).
(k) Government installation where tests are to be conducted (see 4.3.3).
(l) When the Government is to furnish the bakery products for the test (see 4.4).
(m) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(n) When the supplier is to furnish supervision of installation (see 4.3.3 and 6.3).

6.2.2 Contract data requirements. Any data items to be delivered under any contract for the items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 Installation supervision. When specified (see 6.2), supervision of installation shall be supplied by the contractor for ovens to be erected in Continental U.S.A. The contractor shall furnish the services of a competent superintendent who shall be familiar with the construction, erection, operation, maintenance, and repair of the equipment and the required utility connections. Material required for complete installation, other than that specified herein, and all installation labor will be furnished by others. All utilities will be extended and connected to the equipment by others. The superintendent shall supervise the installation of the equipment and all utility connections thereto, using the material furnished as stated. Supervision of utility connections shall include all work from points within 10 feet of the oven. Upon completion of the work, and Government acceptance of the oven, the superintendent shall turn the equipment over to the contracting officer or his representative, in proper operating condition, and shall instruct personnel designated by the contracting officer in the proper operation.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one oven. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

6.5 Recipe. The recipe for cherry pie is standard with Department of Defense. Pies shall be baked in parchment-lined paper pie-plates with metal rims, loaded directly on the oven trays.

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Cherry Pie
(canned - pregelatinized starch)

Yield: 100 Portions (17 Pies) | Each Portion: 1/6 Pie
Pan Size: 9 inch Pie Pan | Temperature: 425°F Oven

Ingredients	Weights	Measures	Method
Pie shells, unbaked		17	1. See recipe cards I-G-2 and I-1.
Cherries, canned, red sour, pitted	25 lb 12 oz	3 gal (4 No. 10 cn)	2. Thoroughly drain cherries; set aside for use in step 5. Reserve juice for use in step 4.
Sugar, granulated Starch, pregelatinized Salt	7 lb 1 lb 4 oz	1 gal 4-1/3 cups 1 tbsp	3. Combine sugar, starch, and salt in dry mixer bowl. Blend thoroughly. Use beater at low speed. Do not whip.
Juice, cherry Food coloring, red (optional)		2 qt 2 tsp	4. Add juice gradually to sugar mixture at low speed. Add red coloring. Scrape bowl once after adding liquid. Continue mixing until smooth.
			5. Carefully fold cherries into thickened mixture.
			6. Pour about 2-3/4 cups filling into each unbaked pie shell. Cover with top crust. Seal edges.
			7. Bake 30 to 35 minutes, or until lightly browned.
			8. Cut 6 wedges per pie.

Pie Crust

Yield: 100 Portions (17-2 Crust Pies) | Each Portion: 1/6 Pie
Pan Size: 9 inch Pie Pan | Temperature:

Ingredients	Weights	Measures	Method
Flour, wheat, hard, sifted 1/	8 lb	8 qt	1. Sift flour into mixer bowl.
Shortening	4 lb 12 oz	2-1/2 qt	2. Add shortening. Mix on low speed about 1/2 minute or until mixture resembles coarse corn meal.

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Pie Crust (cont'd)

Yield: 100 Portions (17-2 Crust Pies)			Each Portion: 1/6 Pie
Pan Size: 9 inch Pie Pan			Temperature:
Ingredients	Weights	Measures	Method
Water, cold (40° to 50°F)		1-1/2 qt	3. Dissolve salt in water and gradually add to flour and shortening mixture. Mix on low speed about 1 minute or until dough is just formed.
Salt	4 oz	6 tbsp	
	4 oz	6 tbsp	4. Chill dough 1 hour or longer for ease in handling.
			5. Follow procedures on recipe cards I-G-1 or I-G-2.

1/ 4 lb 12 oz (4-3/4 qt) sifted hard wheat flour and 3 lb 4 oz (3-3/4 qt) sifted soft wheat flour may be substituted for the 8 lb hard wheat flour in step 1.

Custodians:

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Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7310-0361

Review activities:

Army - MD
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